

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	3 October 2036
Team ID	SWTID-2026-5347
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://docs.servicenow.com/bundle/washingtondc-build-workflows/page/administer/flow-designer/concept/flow-designer.html>

ServiceNow IT Procurement Automation Architecture

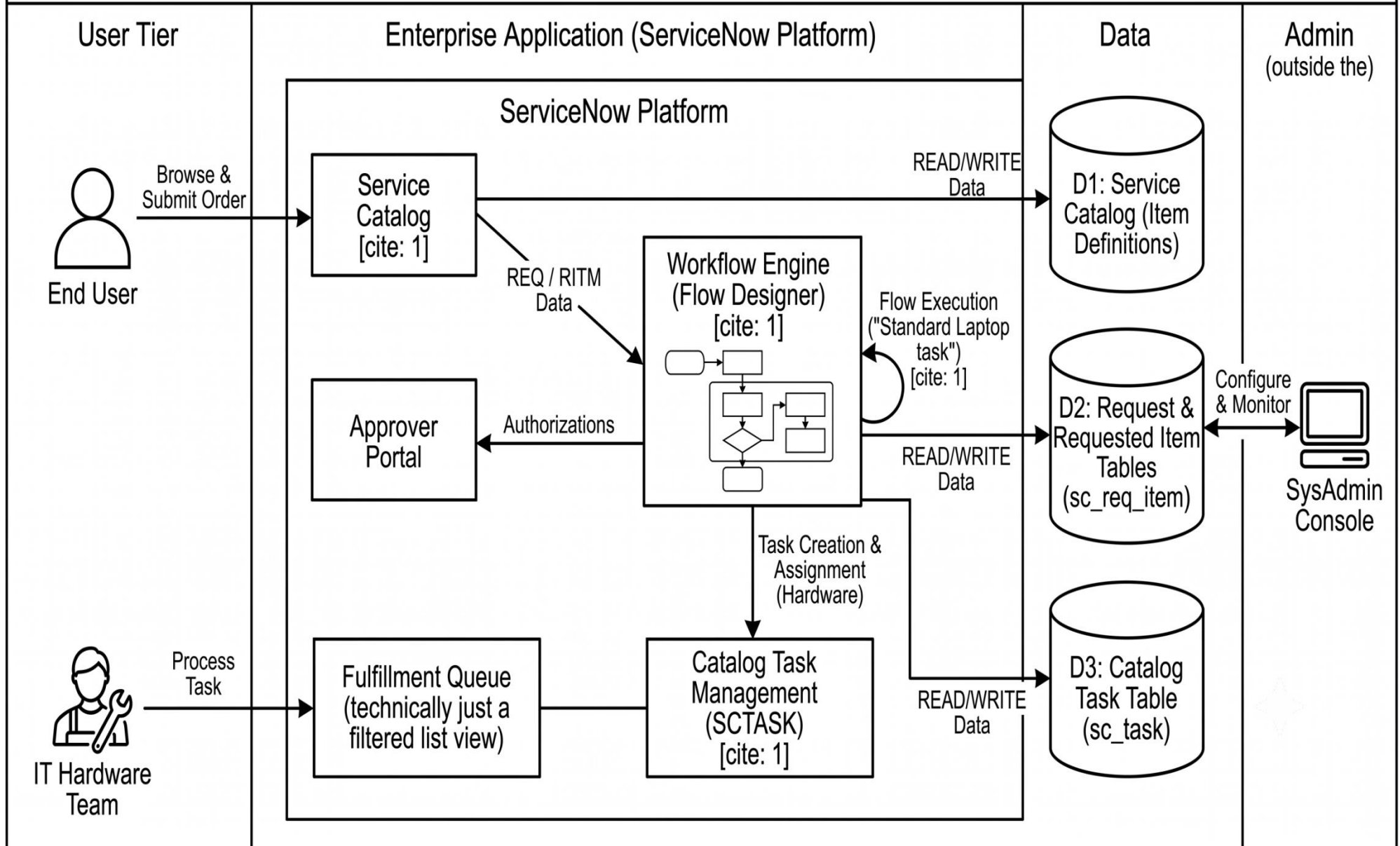


Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	How user interacts with application e.g. Web UI, Mobile App, Chatbot etc. Web portal for browsing hardware catalog items, configuring options, ordering, and reviewing status updates.	ServiceNow Service Portal, HTML5, CSS3, AngularJS, Bootstrap
2.	Application Logic-1	Logic for a process in the application Workflow orchestration engine for triggering automated tasks, condition evaluations, and variable handling without custom code.	ServiceNow Flow Designer
3.	Application Logic-2	Logic for a process in the application Catalog administration and process engine linking standard hardware items to designated fulfillment flows	ServiceNow Service Catalog (Maintain Items engine)
4.	Application Logic-3	Logic for a process in the application Fulfillment task routing, lifecycle updates, and assignment tracking.	ServiceNow Task Management Engine
5.	Database	Data Type, Configurations etc. Normalized schema storing catalog configurations, requests, tasks, and user profiles.	ServiceNow Multi-tenant Data Model (MariaDB/MySQL underlying engine)

6.	Cloud Database	Database Service on Cloud Cloud-hosted dedicated database instances ensuring continuous replication and data persistence.	ServiceNow Advanced High Availability (AHA) Cloud Infrastructure
7.	File Storage	File storage requirements Storage mechanism for system logs, catalog item image icons, hardware spec attachments, and task documentation.	ServiceNow Cloud Document & Database Blob Storage
8.	External API-1	Purpose of External API used in the application Integration point for reading/updating hardware asset configurations and tracking serial numbers in an external CMDB.	ServiceNow Table API / REST API
9.	External API-2	Purpose of External API used in the application Corporate directory service integration to validate requester profiles and reporting lines for approvals.	Single Sign-On (SSO) SAML 2.0 / LDAP Integration
10.	Machine Learning Model	Purpose of Machine Learning Model Smart categorization and automated incident/request deflection for hardware catalog lookups	ServiceNow Predictive Intelligence (NLP/Classification)
11.	Infrastructure (Server / Cloud)	Application Deployment on Local System / Cloud Local Server Configuration: Cloud Server Configuration : Cloud Platform-as-a-Service (PaaS) providing managed hosting, instance isolation, and automatic platform updates.	ServiceNow Enterprise Cloud PaaS

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frontend and client-side web technologies supporting interactive catalog forms and responsive service portal views.	AngularJS, Bootstrap Framework, jQuery
2.	Security Implementations	Role-Based Access Control (RBAC), Access Control Lists (ACLs) restricting task edits to the Hardware group, TLS encryption in transit, and AES-256 at rest.	ServiceNow ACLs, RBAC, TLS 1.3, AES-256
3.	Scalable Architecture	Multi-instance enterprise cloud architecture separating presentation, workflow logic, and data layers to handle concurrent order spikes.	ServiceNow Multi-Instance Cloud Architecture (PaaS)
4.	Availability	High availability guaranteed through active-active multi-datacenter replication and automated failover	ServiceNow Advanced High Availability (AHA) Data Centers

S.No	Characteristics	Description	Technology
		systems.	
5.	Performance	Asynchronous background execution of Flow Designer jobs, edge caching for catalog pages, and index-optimized relational queries.	ServiceNow Flow Engine Async Processing & Global Edge Network

References:

- <https://docs.servicenow.com/bundle/washingtondc-build-workflows/page/administer/flow-designer/concept/flow-designer.html>
- <https://docs.servicenow.com/bundle/washingtondc-build-workflows/page/administer/flow-designer/concept/flow-triggers.html>
- https://docs.servicenow.com/bundle/washingtondc-servicenow-platform/page/product/service-catalog-management/concept/c_ServiceCatalogManagement.html
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- <https://www.servicenow.com/content/dam/servicenow-assets/public/en-us/doc-type/white-paper/advanced-high-availability-architecture.pdf>
- https://developer.servicenow.com/dev.do#!/learn/learning-plans/washingtondc/servicenow_application_developer
- <https://www.axelos.com/certifications/itil-service-management>
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- <https://c4model.com/>[cite: 4]
- <https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>[cite: 4]